Data Structures

Bars

Time bars (1min bars) sample markets at fixed intervals.
Wastes samples during quiet periods, under-samples during active ones.

Tick Imbalance Bars (TIBs)

Create sequence of bars $\{b_t\}_{t=1...T}$ where

$$b_t = \begin{cases} b_{t-1} & \text{if } \Delta p_t = 0 \\ \frac{|\Delta p_t|}{\Delta p_t} & \text{if } \Delta p_t \neq 0 \end{cases}$$

p_t, price at t
$b_t \in \{1; -1\}$

Tick imbalance measured by: $\theta_T = \sum_{t=1}^{T} b_t$ (sum of 1 or -1)
If buys dominate θ_T large and positive.
If sells dominate θ_T large and negative.

Setting expectation $E_0[\theta_T] = E_0[T] \times (P(b_t=1) - P(b_t=-1))$
$= E_0[T] \times (2P(b_t=1) - 1)$
The bar closes when $|\theta_T| \geq |E_0[\theta_T]|$

- Informed / One-side trading → θ_T grows fast, bars close quickly
- Balanced / noise trading – θ_T near zero, bars stay open longer

Volumes / Dollar Imbalance Bars (VIBs / DIBs)

Imbalance θ_T now volume / dollar weighted

$\theta_T = \sum^{T} b_t \times v_t$ (or $p_t \times$ shares)

Expectation:

$v^+ = P(b_t=1) \times E_0[v_t \mid b_t=1]$
$v^- = P(b_t=-1) \times E_0[v_t \mid b_t=-1]$
↳ $E_0[v_t] = v^+ + v^-$
$= E_0[T] \times \underbrace{(v^+ - v^-)}_{\text{net directional bias in volume}}$

Volume / Dollar Runs Bars (VRBs / DRBs)

θ_T measures the dominant run: $\theta_T = \max(\sum b_t = 1, \sum b_t = -1)$
Set $E_0[\theta_T] = E_0[T] \times \max(P[b_t=1], P[b_t=-1])$
↳ since $P[b_t=1] + P[b_t=-1] = 1$, $\max(...)$ is always ≥ 0.5.
↳ in balanced market close to 0.5, in directed close to 1.
Cancel point is different:
$\theta_T \geq E_0[T] \times \max(P[b_t=1], P[b_t=-1])$

⚠ TRBs measure how much trading is happening on the dominant side, TIBs measure imbalance.

Event Driven Sampling

Random sampling:
- Most bars nothing interesting
- Model trains on noise + signal mixed
- Result: low accuracy, learned noise

Selective sampling:
- Only bars where catalytic conditions exist
- Model trains on periods where features are informative
- Result: higher accuracy, learns signal

CUSUM Filter

Upward CUSUM: $S_t^+ = \max\{0, S_{t-1}^+ + y_t - y_{t-1}\}$
where, y_t: current observation
$y_t - y_{t-1}$: surprise - deviation this period
S_{t-1} + surprise: cumulative surprise so far
$\max\{0, ...\}$: floor at zero, reset when cum. surprise negative
Downward CUSUM: $S_t^- = \min\{0, S_{t-1}^- + y_t - y_{t-1}\}$

Filters for cumulative deviation superior to a threshold.
Applied on stock prices allows for selective sampling of points.